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Review Article

Next-generation approaches in autoimmune encephalitis diagnosis: A multi-omics perspective☆

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ABSTRACT

Autoimmune encephalitis (AE) is a group of autoantibody-mediated inflammatory disorders of the central nervous system (CNS) that often present with cognitive deficits, behavioral abnormalities, and seizures. The subtypical variability of AE calls for individualized treatments based on specific antibodies and clinical manifestations. Early and accurate diagnosis and standardized treatment are essential to improve prognosis and promote neurological recovery. Currently, clinical diagnosis of AE relies on the detection of specific antibodies in serum and cerebrospinal fluid (CSF). However, there is usually a delay in this approach, leading to late initiation of treatment, which may result in patients' disease progression yet still not effectively treated. For example, for some specific subtypes of AE, existing antibody tests may not capture all potential markers in time, making early diagnosis and subtyping challenging. In addition, certain AE patients have low antibody levels that make it difficult to confirm the diagnosis with conventional assays, resulting in a delayed diagnosis. Therefore, there is an urgent need for new methods to improve the efficiency of differential diagnosis based on the sensitivity, specificity, or accessibility issues faced by antibody-based diagnosis.

With the help of multi-omics technologies, potential biomarkers of AE can be identified through genomic, proteomic, and metabolomic analyses, leading to improved diagnosis. Multi-omics technologies have demonstrated great potential in the diagnosis and treatment of AE, especially in revealing new markers and improving early diagnosis, but the limitations of technological complexity high cost, and the problem of clinical applicability still need to be overcome. In the future, with technological advances and cost reductions, multi-omics approaches are expected to become an important tool for the early diagnosis of AE and promote more accurate and timely treatment.

1. Introduction

Autoimmune encephalitis (AE) comprises a heterogeneous group of inflammatory diseases of the central nervous system mediated by autoimmune mechanisms. It presents as a series of neurologic dysfunctions caused by diffuse or multiple inflammatory lesions in the brain parenchyma. The pathologic changes are predominantly gray matter with neuronal involvement, but may also involve white matter and blood vessels. Common clinical features of AE include behavioral changes, psychiatric abnormality, seizures, memory and cognitive deficits, autonomic abnormalities, and decreased levels of consciousness (Dalmau et al., 2018; Nissen et al., 2020), whose typical symptoms reflect dysfunction of the limbic structures of the brain. Different subtypes of AE may have different clinical presentations. A prospective,

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multicenter, population-based study has now shown that autoimmune diseases are the third most common cause of encephalitis, after infections (usually viral) and acute disseminated encephalomyelitis (usually postinfectious diseases) (Granerod et al., 2010). A study by a center focusing exclusively on the epidemiology of encephalitis showed that the frequency of the most common type of autoimmune encephalitis, that is, the type with anti-*N*-methyl-D-aspartate receptor (NMDAR) antibodies, exceeded the frequency of any individual viral cause of encephalitis in young people (Gable et al., 2012).

Early diagnosis of AE is challenged by the uncertainty of antibody test results and the diversity of clinical presentations. Existing antibody detection methods suffer from low sensitivity and delayed detection, leading to missed or delayed diagnosis in some patients. In addition, due to the atypical nature of the clinical manifestations of AE, diagnosis often relies on multiple tests and the exclusion of other possible diseases, which further increases the complexity of diagnosis. In terms of treatment, as the current treatment options for AE are not comprehensive and effective, prognosis is difficult to determine and treatment of recurrence is tricky. Although immunotherapy can provide some degree of symptomatic relief, many patients are still at risk of long-term relapse after treatment. Therefore, the discovery of more effective biomarkers is clinically important to improve diagnosis, treatment, and prognosis.

Multi-omics studies provide a multi-dimensional perspective that enables the identification of significantly differentially expressed biomarkers at different levels such as genes, proteins, and metabolites. This approach helps to gain a deeper understanding of the interactions of different factors under specific conditions, thereby revealing potential pathogenic mechanisms of AE and identifying possible targets for clinical intervention or drug therapy. However, despite the theoretical potential of multi-omics for a wide range of applications, there are certain limitations in practice, such as high cost, complexity of data processing, and lack of sufficient validation studies. These factors limit the widespread clinical application of multi-omics approaches, especially in the study of complex diseases such as autoimmune encephalitis (AE). In previous AE studies, multi-omics has helped identify some promising biomarkers that provide new answers for diagnosis and monitoring of treatment response in AE (Li et al., 2025a; Li et al., 2025b), but its translation into clinical practice still faces many challenges.

This article briefly describes the different AE antibody assays and other detection tools and summarizes the progress of multi-omics research based on AE biomarkers. Although these studies provide potential biomarkers for the early diagnosis of AE, how to effectively apply these markers in clinical practice remains a challenge yet to be solved. Future studies need to make progress in overcoming the limitations of multi-omics technologies, expanding sample sizes, increasing validation studies, and optimizing clinical translational pathways to achieve the practical application of multi-omics approaches in the diagnosis and treatment of AE.

2. Progress in the study of autoimmune encephalitis

Since the discovery of anti-*N*-methyl-D-aspartate receptor (NMDAR) (Dalmau et al., 2007) encephalitis in 2007, a series of autoantibodies against neuronal cell-surface or synaptic proteins have been discovered. A variety of subtypes of encephalitis have been successively reported by cell-based detection techniques as well as by relevant immunoassays, including some individual cases of rare encephalitis subtypes. Unlike classical paraneoplastic limbic encephalitis, the target antigens of AE are located on the surface of neuronal cells or intracellularly, causing relatively reversible neuronal dysfunction mainly through humoral immune mechanisms (Graus et al., 2016; Probst et al., 2014; Gresa-Arribas et al., 2016). Currently, AE accounts for 10 % to 20 % of encephalitis cases, with anti-NMDAR encephalitis being the most common, accounting for 54 % to 80 % of AE cases (Ren et al., 2021), followed by anti-leucine-rich glioma-inactivating protein 1 (LGI1) antibody-associated encephalitis and γ-amino butyric acid type B receptor ($GABA_BR$) antibody-associated encephalitis (Davies et al., 2010; Guan et al., 2016; Suh-Lailam et al., 2013). In addition, some different antibody-positive rare types of encephalitis are included.

Depending on the site of the presence of anti-neuronal antibodies, AE can be classified into two types: (1) AE with antibodies directed against extracellular neuronal antigens, which may be centered on either unrecognized antibodies or pathogenesis mediated primarily by T cells (Dalmau et al., 2018); (2) AE is associated with noncellular antibodies directed against intracellular neuronal antigens and is usually strongly associated with paraneoplastic syndromes (Bien et al., 2012). Although different subtypes of AE encephalitis have similarities in clinical manifestations and diagnostic means, their specific diagnostic and therapeutic approaches vary considerably due to differences in their antibodies and differences in pathogenesis. Table 1 summarizes the different subtypes of AE encephalitis. Common prodromal symptoms of AE include fever and headache, and the main symptoms are characterized by abnormal mental behavior, cognitive deficits, memory loss of recent events, seizures, speech disorders, movement disorders, involuntary movements, decreased level of consciousness with coma, and autonomic dysfunction.

2.1. Antibody testing for autoimmune encephalitis

Currently, the diagnosis of AE is based on a positive autoantibody test as one of the final diagnostic criteria, which is confirmed by combining the patient's clinical manifestations, cerebrospinal fluid (CSF) examination, neuroimaging, EEG, and other auxiliary examinations. Currently, antibody detection methods for AE are mainly based on serum or cerebrospinal fluid samples, and commonly used methods include Western blotting (WB), enzyme-linked immunosorbent assay (ELISA), and immunofluorescence assay (IFA). These methods can be further categorized into cell-based assays (CBA) and tissue-based assays (TBA) based on the substrates they detect. Table 2 summarizes the characteristics of several antigen assays.

Despite the widespread use of these methods in clinical practice, there are limitations. First, a positive antibody test result does not represent a definitive diagnosis of disease, and many patients may still have AE even if they do not show a positive antibody result (Ricken et al., 2018; Lascano et al., 2019). Therefore, a negative result on an antibody test cannot completely rule out the possibility of AE. Second, available antibody tests do not accurately differentiate between different types of encephalitis, which may be due to antibody epitopes that cannot be captured by existing ELISA or IHC techniques. More importantly, antibody tests often take several days to produce results, making them unsuitable as early diagnostic tools for AE. In patients with acute or acute-subacute stages, delays in antibody testing may miss the optimal time for treatment.

To overcome these limitations, future research needs to focus on improving the sensitivity and specificity of antibody detection. This includes the development of new assay platforms, such as multiplex or combined assay strategies, utilizing more sensitive assays that incorporate high-throughput technologies and biomarkers that may be able to improve diagnostic accuracy. In addition, in-depth studies on the correlation between antibody titers and clinical symptoms may help to establish more accurate prognostic assessment criteria, thereby enhancing the effectiveness of personalized treatment. In summary, although antibody testing for autoimmune encephalitis provides important information in clinical diagnosis, its limitations still cannot be ignored. Future studies need to further improve the antibody detection technology and combine it with other clinical tests to provide more accurate diagnosis and treatment options for patients.

Currently, antibody-based diagnostic techniques are mostly used in clinical practice, and in the face of their limitations, existing research is committed to realizing innovations and breakthroughs in emerging diagnostic technologies. For example, Single-cell secretome sequencing technology (SEC-seq) can isolate plasma cells from patients'

Table 1
Different subtypes of AE encephalitis.

Antigen	Syndrome	Risk factor	Clinical symptom	Diagnostic assay
Antibodies against cell surface synaptic receptors				
NMDAR (Dalmau et al., 2011; Dalmau et al., 2008; Shen et al., 2025)	Anti-NMDA receptor encephalitis	More women than men; ovarian teratoma found in25–40 % of women	Mental behavioral abnormalities or cognitive impairment, seizures, motor aphasia	Cell-based assay
$GABA_AR$ (Petit-Pedrol et al., 2014; Spatola et al., 2017; Braczkowski et al., 2023)	encephalitis	40 % thymoma; can be secondary to herpes simplex virus encephalitis	refractory status epilepticus, memory deficits, mental disorders, disturbances of consciousness	Cell-based assay
$GABA_BR$ (Lancaster et al., 2010)	Limbic encephalitis	50 % small cell carcinoma	Severe and refractory seizures can be combined with speech disorders, sleep disorders, and cerebellar ataxia.	Cell-based assay
AMPAR (Lai et al., 2009; Joubert et al., 2015; Laurido-Soto et al., 2019; Wang et al., 2025)	Limbic encephalitis	65 % small cell lung cancer or thymoma	Mainly presenting as limbic encephalitis, but can also present as simple amnesia or even fulminant severe encephalitis, and olfactory hallucinations.	Cell-based assay
mGluR5 (Braczkowski et al., 2023; Lancaster et al., 2011; Spatola et al., 2018; Guo et al., 2023)	encephalitis	70 % combined Hodgkin's lymphoma, rarely small cell lung cancer	Cognitive impairment, seizures, headache, low-grade fever, respiratory symptoms, sleep disturbances, decreased level of consciousness, movement disorders	Cell-based assay
D_2R (Dale et al., 2012)	Basal ganglia encephalitis	No tumor-related	Movement disorders are characterized by Parkinson's disease, dystonia, and chorea.	Cell-based assay
Antibodies against other cell surface proteins				
LGI1 (Irani et al., 2013; Gadoth et al., 2017; Lai et al., 2010; Antonio et al., 2025)	Limbic encephalitis	5–10 % thymic tumors	Episodes of facial and arm dystonia, paroxysmal dizziness episodes, and isolated retrograde amnesia	Cell-based assay
CASPR2 (Antonio et al., 2025; Hartung et al., 2023; Carvelli et al., 2023; Irani et al., 2010)	Morvan's syndrome or limbic encephalitis	20–50 % Thymoma	Insomnia, autonomic dysfunction, ataxia, peripheral nerve hyperexcitability, and associated neuropathic pain, isolated retrograde amnesia.	Cell-based assay
IgLON5 (Sabater et al., 2014)	Encephalopathy with sleep disorders	HLA-DRB1*1001 and/or HLA-DQB1*0501 anomalies	Sleep disorders and movement disorders	Cell-based assay
DPPX (Hara et al., 2017; Boronat et al., 2013)	Encephalitis with diarrhea	<10 % of patients with lymphoma	Prodromal symptoms of severe diarrhea and weight loss	Cell-based assay
Antibodies against intracellular synaptic antigens				
GAD (Malter et al., 2010)	Limbic encephalitis	25 % Thymoma, small cell lung cancer	Epileptic seizures, proximate memory disturbances, and mental behavioral abnormalities	Radioimmunoassay
Amphiphysin (Pittock et al., 2005; Moon et al., 2014; Murinson and Guarnaccia, 2008)	Limbic encephalitis	Combined small-cell lung and breast cancer	Epilepsy, memory disorders, and other borderline system disorders	ELISA
Antibodies against intracellular antigens				
Hu(ANN A-1) (Alamowitch et al., 1997)	Limbic encephalitis	95 % small cell carcinoma	Epilepsy, ictal speech difficulties, cerebellar ataxia	Western blot
CV2/CRMP5 (Rogemond and Honnorat, 2000; Honnorat et al., 2009)	Limbic encephalitis	>80 % small cell lung cancer and thymoma	Chorea, pseudo-intestinal obstruction, and myelopathy can occur	immunofluorescence assay
Ma2 (Dalmau et al., 2004)	Limbic encephalitis , mesolithic encephalitis	95 % testicular spermatogonial cell tumors	Can be secondary to episodic sleeping sickness	Western blot

Amphiphysin or CV2 (CRMP5) antibodies instead of Hu antibodies in a few patients with limbic encephalitis and small-cell lung carcinoma.

cerebrospinal fluid and directly sequence their secretory antibodies, circumventing the problem of low abundance of serum antibodies (Cheng et al., 2023). In addition, the development of new biomarkers holds promise for the diagnosis of AE.

In recent years, multi-omics research has provided a promising avenue. Multi-omics screening of new biomarkers based on disease cohorts to further expand and improve the spectrum of humoral markers for AE is expected to be used for early diagnosis, disease course monitoring, and efficacy assessment of AE.

3. Multi-omics study identifies potential biomarkers

3.1. GWAS for identifying sensitive gene loci

Genome-wide association Studies (GWAS) are capable of searching for genetic factors associated with complex diseases through genome-wide high-density genetic marker (e.g., SNPs or CNVs, etc.) typing of large-scale population DNA samples, which can comprehensively reveal the genetic genes associated with disease onset, progression, and treatment. GWAS is a powerful tool for studying the genetic factors and possible pathogenic mechanisms of autoimmune encephalitis and can identify sensitive genes for AE to assist in the early diagnosis of AE.

In previous studies, several risk loci with significant genome-wide variation have been identified, including the LRRK1 gene on chromosome 15 and the ACP2 and NR1H3 genes on chromosome 11 associated with anti-NMDAR encephalitis (Tietz et al., 2021), the DCLK2 gene on chromosome 4 is potentially associated with anti-LGI1 encephalitis (Mueller et al., 2018). The LRRK1 gene is expressed in B cells and monocytes and plays a role in the immune system (Thévenet et al., 2011). Its defective mice exhibit altered B-cell development, failure to

Table 2
Characterization of the test for several different antigens.

Diagnostic assay	Appliance	Challenge	Improvements	References
Tissue-based assay(TBA)	Commonly used for the primary screening of multiple AE antibodies, discovery of novel unknown AE antibodies.	Detects a wide range of antigens but does not recognize the exact molecular target of the autoantibody	The detection accuracy of TBA can be improved by increasing the sensitivity of two-photon microscopy to enhance its ability to recognize antigens. In addition, changing the tissue section into a live carrier, combined with two-photon microscopy imaging technology, can achieve more accurate in vivo observation and enhance target recognition.	(Liu et al., 2019; van Coevorden-Hameete et al., 2016)
Cell-based assay (CBA)	Antibodies to cell surface or synaptic proteins	Detects overexpression of only one antigenic target in cells; highly sensitive, specific, and reliable results	It can be improved by Multiplex CBA technology, which provides a more comprehensive analysis of the antibody response by labeling magnetic beads with different antibodies and enabling multiplex detection using flow cytometry.	(van Coevorden-Hameete et al., 2016; Gu et al., 2019a) (Anderson, 2011)
ELISA	Intracellular AE antigen detection	High sensitivity and strong quantitative ability, but false negatives and false positives may occur.	eSimoa (Enhanced Single Molecule Immuno-Assay) is a new type of high-sensitivity immunomolecular analyzer, mainly compared with the traditional ELISA, eSimoa is more sensitive and efficient for the detection of molecules at low concentrations in biological samples.	(Ricken et al., 2018; Willison et al., 2000) (Wang et al., 2018)
Western blot (WB)	Intracellular AE antigen detection	With low detection sensitivity, false negatives may occur	JESS Instruments' fully automated WB analysis ensures that the best antibody signal is obtained for each experiment by automatically adjusting the experimental conditions to more accurately recognize the weak reaction between the antibody and the target antigen.	(Ricken et al., 2018; Willison et al., 2000) (Sharifi et al., 2019)
Radio immune assay(RIA)	Detection of antibodies against channel complexes such as GAD-Abs and VGKC complexes	High sensitivity and specificity, but inability to distinguish between antibodies directed against different channel components that may be of clinical relevance	RIA can be replaced by adopting modern PET technologies such as TSPO-PET, which can provide higher resolution and specificity to help differentiate between different antibody components of clinical relevance and improve diagnostic accuracy.	(van Coevorden-Hameete et al., 2016; Motomura et al., 1995) (Kraus et al., 2018)

produce IgG3 antibodies in response to non-T-cell-dependent antigens, and defects in B-cell receptor-stimulated proliferation and survival (Morimoto et al., 2016). Able to demonstrate that neuroimmune mechanisms also play an important role in the pathogenesis and progression of AE. The DCLK2 gene is predominantly expressed in multiple brain regions (GTEx) and promotes neuronal development and survival, and its absence in mice leads to hippocampal dysplasia and epilepsy (Kerjan et al., 2009; Nawabi et al., 2015). GWAS studies of different subtypes of AE can identify sensitive gene loci and reveal the pathogenesis of different subtypes of AE at the genetic level. There have also been studies looking at the association of HLA polymorphisms with AE. Stefanie et al. found a significant association between anti-NMDAR encephalitis and the MHCI allele HLA-B * 07:02, which was stronger in a subgroup of patients with later onset, but absent in a subgroup with earlier onset (Mueller et al., 2018).

The discovery of AE susceptibility genes by GWAS and the ability to differentiate between different subtypes of AE encephalitis are informative in the early diagnosis of AE.

3.2. Non-coding RNAs can serve as potential biomarkers

The transcription process of synthesizing RNA using DNA as a template is the first step in gene expression and a key link in the regulation of gene expression. The RNA molecules formed by transcription of the human genome include coding RNA and non-coding RNA. mRNA that can be translated into proteins only accounts for about 40 % of the entire transcriptome, and non-coding RNAs such as microRNAs and circRNAs can regulate the translation and expression of mRNAs, and most of them are inhibitory. J.S. Jun et al. (Ozturk, 2017) investigated changes in the expression profiles of mRNAs in an in vitro model of NMDAR encephalitis. 17,509 mRNAs were analyzed, of which 134 mRNAs were upregulated and 57 mRNAs were downregulated. Downstream pathway analysis revealed that the most abundant functional category of genes was “cell differentiation”, which included 38 mRNAs. Another study used microarray analysis to assess mRNA expression profiles in blood leukocytes from patients with anti-NMDA receptor encephalitis and healthy controls and found that 719 mRNAs were dysregulated in patients with anti-NMDA receptor encephalitis (Guo et al., 2020a). All of the above findings show that dysregulated mRNAs are more common in AE patients and are potentially valuable to study.

Much current research focuses on the possible role of small noncoding RNAs in the pathogenesis of AE and their potential as biomarkers of AE. MicroRNAs are a group of small endogenous noncoding RNA molecules that regulate up to one-third of protein-coding genes. Most miRNAs are inhibitory in their role in gene regulation, either directly binding and preventing translation of target mRNAs or indirectly inducing early decay and degradation of mRNAs. A previous study used an in vitro model to identify four miRNAs, miR-139-3p, miR-6216, miR-135a-3p, and miR-465-5, which were differentially expressed in anti-NMDAR encephalitis (Jun et al., n.d.). Zhang J et al. found that let-7a, let-7b, let-7d, and let-7f were down-regulated in anti-NMDAR encephalitis, and among them, microRNA let-7b has the potential to be used as a biomarker for anti-NMDAR encephalitis (Zhang et al., 2015). Other studies have also demonstrated that the miRNA let-7 family is associated with anti-NMDAR encephalitis (Wang, 2019). Let-7 was one of the first miRNAs to be identified from its first (Alevizos and Illei, 2010), with let-7e shown to be associated with autoimmune encephalomyelitis (Cao et al., 2015). Li Y et al. (Li et al., 2022a) performed pathway enrichment of differentially expressed miRNA profiles by isolating purified exosomes from cerebrospinal fluid and found that differentially expressed miRNAs were mostly enriched in cancer-related pathways, including endometrial cancer, p53 signaling pathway, and non-small cell lung cancer. In addition, previous studies have found that underlying tumors are present in patients with AE and can stimulate autoantibody production (Tanaka et al., 2020; Tirthani et al., 2023).

In addition to miRNAs, circular RNAs (circRNAs) have been identified as potential biomarkers for AE encephalitis. Jiamin Guo et al. used microarray analysis to study the role of circRNA in anti-NMDA encephalitis in children (Guo et al., 2020a). They found 1196 circRNAs dysregulated in patients with anti-NMDA receptor encephalitis, and further bioinformatics analysis showed that the two most abundant KEGG pathway terms for dysregulated circRNAs were endocytosis and lysosomes, both of which are associated with receptor internalization. According to previous studies, the possible pathogenic mechanism of NMDAR is receptor internalization (Hughes et al., 2010; Moscato et al., 2014), and thus, dysregulated circRNAs may become new therapeutic targets with a potential role in predicting the progression of AE encephalitis as well as in prognostic therapy.

3.3. Biomarkers identified by proteomics

Proteomics, an unbiased method for assessing protein abundance in cerebrospinal fluid, has the potential to identify novel biomarkers, facilitate early diagnosis, and promote outcome prediction, which in turn can influence treatment options (Li et al., 2022b; Hörste et al., 2020). Cerebrospinal fluid proteomics studies for other autoimmune diseases of the central nervous system (e.g., multiple sclerosis) have led to several studies that have revealed new insights into the pathogenesis of the diseases and helped to identify novel cerebrospinal fluid biomarkers (Teunissen et al., 2015; Khalil et al., 2018). Potential biomarkers identified in current studies for AE treatment response and outcome include CXCL13, NLRP3, NSE, S100B, Nf-L, and others.

CXCL13, a B-cell-attracting chemokine, is mildly elevated in autoimmune diseases such as multiple sclerosis and optic neuromyelitis optica (Zhong et al., 2011; Khademi et al., 2011). Cerebrospinal fluid CXCL13 levels correlate better with the presence of plasma cells or plasmablasts in the CNS than other B-cell-attracting chemokines (Kowarik et al., 2012). One study found that 70 % of patients with early-onset anti-NMDAR encephalitis had elevated CSF CXCL13 concentrations, which correlated with intrathecal NMDAR antibody synthesis. Chronic or secondary elevations in CXCL13 were associated with a limited response to treatment and relapse (Leypoldt et al., 2015).

NLRP3 inflammasome is a potent innate inflammatory mediator that activates IL-1β and induces T-helper cell migration into the CNS (Peng et al., 2019), which can activate the innate immune system and lead to tissue damage (Kelley et al., 2019). Cerebrospinal fluid NLRP3 inflammatory vesicles and their downstream cytokines were found to be significantly increased in patients with anti-NMDAR encephalitis in the acute phase, whereas their respective levels were reduced in the remission phase. Cerebrospinal fluid NLRP3 inflammatory vesicles and their downstream cytokines were found to be significantly increased in patients with anti-NMDAR encephalitis in the acute phase, whereas their respective levels were reduced in the remission phase (Peng et al., 2019). IL-6 and IL-17 were elevated in patients with anti-NMDAR encephalitis, while the levels of cerebrospinal fluid NLRP3 inflammatory vesicles were positively correlated with either IL-6 or IL-17, suggesting that co-activation of IL-17/IL-6 in anti-NMDAR encephalitis may be the pathogenesis of anti-NMDAR encephalitis (Peng et al., 2019).

Liu Et al. in their experiments found that patients with anti-NMDAR encephalitis had significantly higher cerebrospinal fluid NSE and S100B concentrations than controls (Liu et al., 2018). Neuron-specific enolase (NSE) is a dimeric enzyme formed in neurons and neuroendocrine cells with subordinate units α-γ or γ-γ and belongs to the class of hydrolases (Schmechel et al., 1978; Isgrò et al., 2015). S100 Calcium-binding protein B (S100B) belongs to the calcium-binding protein family. It is expressed predominantly in astrocytes and to some extent in a small fraction of oligodendrocytes (Gos et al., 2013; Hachem et al., 2005). Studies have shown that when CNS injury is accompanied by nerve tissue and cellular damage, CNS structural proteins (including NSE and S100B) are released from cells and their extracellular concentrations are increased, including in cerebrospinal fluid and blood (Persson et al., 1987; Kleine et al., 2003; Zetterberg et al., 2013).

Biomarkers can also be applied to discriminate patients with encephalitis at different periods of onset, and cerebrospinal fluid Nf-L levels are significantly elevated in the acute phase of patients with anti-NMDAR encephalitis and positively correlate with disease severity (Li et al., 2019), so that Nf-L is considered a potential clinical outcome and a useful indicator of prognosis. Neurofilaments (Nf) are a series of highly specific scaffolding proteins of neurons, the absence of which affects the morphological integrity of axons and the efficient conduction of nerve impulses (Li et al., 2019). Under conditions of neurodegeneration or neuronal cell death, the level of Nf increases in cerebrospinal fluid (CSF) or serum, and the level of change is the level at which the degree of neurodegeneration or neuronal death can be determined (Petzold, 2005; Gaiottino et al., 2013; Petzold et al., 2011).

In addition to differentially expressed marker proteins, some studies have also attempted to capture changes in post-translational protein modifications. Räuber S et al. found reduced levels of GSTP1 in NMDAR-AE and LGI1-AE, suggesting that dysregulated oxidative stress is a component of the complex pathophysiology of AE (Räuber et al., 2023). GSTP1 is a protein that prevents neurodegeneration (Sun et al., 2011) involved in glutathione metabolism, mediating antitoxic and antioxidant effects (Perricone et al., 2009). They found that several proteins associated with the cellular stress response were elevated to varying degrees in all AE isoforms compared to HC and that some other proteins protected against oxidative stress that were lower (Räuber et al., 2023). Increased oxidative stress, in turn, leads to elevated oxidized protein modifications. These modifications may be useful biomarkers for assessing oxidative stress in physiological and pathological conditions (Cai and Yan, 2013). It was also found that in GAD-AE, ADAMTSL2 (an extracellular matrix protein (Rypdal et al., 2021)) and PLCG1 (a protein associated with the oxidative stress response (Bai et al., 2002)) are one of the unique proteins with the highest differential abundance (Räuber et al., 2023).

3.4. Metabolomics

3.4.1. Differential analysis of biochemical metabolic pathways

Metabolomics is a new approach to elucidating pathomechanisms and biomarkers and to assessing the environmental impact of disease. The metabolome reflects downstream changes in genomic, transcriptomic, and proteomic expression and has a powerful ability to recognize small changes in gene expression and protein translation that are difficult to detect in other histological studies (Rinschen et al., 2019).

Sensitizing factors and pathogenesis of AE were investigated by studying changes in serum/plasma metabolic profiles and fecal metabolites in patients with AE, and thus identifying changes in information pathways. Several studies have shown that metabolic abnormalities in patients with NMDAR encephalitis primarily include: (1) Cancer pathways were significantly elevated in both choline metabolism and bile secretion pathways (Gong et al., 2022); (2) Altered metabolism of multiple amino acids, including highly enriched tryptophan and glutathione metabolism (Chen et al., 2020); (3) Lipopolysaccharide (LPS) synthesis was increased and coincided with increased serum LPS levels. LPS enables antibodies against NMDAR encephalitis to enter the brain by inducing the production of pro-inflammatory cytokines (Chen et al., 2020); (4) Decrease in short-chain fatty acids (SCFA) increases damage to the intestinal barrier (Kelly et al., 2015). In validating intestinal mucosal injury, Chen et al. found that serum levels of two chemical markers, D-LAC and DAO, were significantly elevated in patients with anti-NMDAR encephalitis (Chen et al., 2020), and that both markers are usually at higher levels during intestinal barrier disruption (Ji et al., 2018).

In addition to general metabolites, several studies have found dysregulation of cytokines in patients with AE. Cytokines are key drivers of inflammation and tissue damage in various immune-related diseases (Moudgil, 2015). As quantifiable indicators, cytokines can be used as biomarkers for AE diagnosis. Gong et al. found that some cytokines were significantly higher in the patient group than in HC, including IFN-β, IFN-γ, IL-3, IL-6, IL-1, IL-7, IL-18, IL-33, and TNF-α. These cytokines were elevated in patients with severe status, increased modestly in patients with moderate status, and suppressed in patients with HC, further confirming the correlation between cytokines and graded levels of disease severity (Gong et al., 2022).

In addition, Th17 cells are important immune cells involved in the pathogenesis of AE, acting as an inflammatory Th subpopulation involved in intrathecal synthesis and B cell activation (Zhang et al., 2022). Th17 cell chemotaxis, growth, and differentiation are regulated by a variety of cytokines and inflammatory molecules and can induce the production of pro-inflammatory factors. Chaosheng et al. demonstrated that Th17 cells accumulate in the CNS of AE patients (Zeng et al., 2018). The RORγt−/-Th17-deficient mouse model also exhibits reduced regulatory T cell numbers and attenuated microglia activation during post-infectious basal ganglia encephalitis (Platt et al., 2020). These findings emphasize the critical role of Th17 cells in the activation of AE.

Interestingly, the above studies found a significant inflammatory factor storm in patients with AE, as evidenced by dysregulation of cytokines associated with inflammatory regressions and an abnormal accumulation of immune cells Th17 cells. With increasing degree and duration of inflammation, the vascular blood-brain barrier becomes more permeable to solutes, with increased lymphocyte transport and infiltration by innate immune cells; occasionally, endothelial cell damage occurs. This, in turn, damages the blood-brain barrier and disturbs normal neuronal function (Galea, 2021; Rust et al., 2025). In addition, Xiao J et al. found antigen presentation and bidirectional activation between microglia and Th17 cells in an experimental autoimmune encephalomyelitis (EAE) model. This process is dependent on MHC-II, proinflammatory cytokines, and inflammatory chemokines, as well as the STING→NF-κB pathway in microglia and effector cytokines produced by Th17 cells, which drive disease progression in EAE (Xiao et al., 2025). Thus, in the metabolomics of AE, the aberrant expression of inflammatory factors interacts with the accumulation of Th17 cells to further exacerbate the inflammatory response, leading to microglia loss of function and disruption of the blood-brain barrier.

3.4.2. Metabolites associated with gut microbes

There is evidence that gut microbes can influence brain function through the gut-brain axis (GMA) (Hsiao et al., 2013) and that disturbances in the gut microbiota or GMA may contribute to CNS autoimmunity and neuropsychiatric manifestations (Chen et al., 2020). Several studies have focused on the role of gut microbes in neurodegenerative diseases such as multiple sclerosis, so there is a potential for gut microbes in the pathogenesis of AE. The use of metagenomics sequencing to characterize the compositional and functional differences of the gut microbiome can lead to discoveries for prognostic and therapeutic determination of AE.

Recent studies have identified significant ecological dysregulation of the gut microbiome in patients with different subtypes of AE encephalitis, including anti-NMDAR encephalitis and anti-LGI1 encephalitis (Gong et al., 2022; Chen et al., 2020; Ma et al., 2020). There are significant changes in the numbers and proportions of several important flora, and their downstream metabolic pathways are also associated with the clinical manifestations of AE. Reduced flora detected by 16sRNA sequencing or metagenomic sequencing included *Clostridium difficile* and Boosters (tapering from remission to active patients), Prevotella_6, *Bifidobacterium bifidum*, *E. faecalis* (Ma et al., 2020), and other short-chain fatty acid (SCFA)-producing bacteria (Gong et al., 2019). In addition, bacteria that have a role in reducing inflammation, such as Fusobacterium, anaerobes, Chironobacter, rumenococci, and butyric acidococci, were also reduced (Gong et al., 2022; Chen et al., 2020).

Enriched flora include streptococci, *E. coli* (Gong et al., 2019) microporous bacteria, dorea, clostridia (Chen et al., 2020) Enterococci, and Salmonella (Gong et al., 2022) which are usually conditionally pathogenic or pathogenic flora. Someflora associated with movement disorders, psychiatric disorders, and pro-inflammatory effects, such as Sphingomonas, anaerobes, Vibrio succinicus, Clostridium, and SMB53 spp. also showed a significant increase in (Ma et al., 2020).

Metabolomics studies can identify end products of microbial metabolism through serum metabolite profiles and fecal assays, which are functionally more important than specific bacterial species (Holmes et al., 2012; Agus et al., 2018). Symbiotic microbiomes modulate cytokine-mediated immune responses (Fatkhullina et al., 2018; Guo et al., 2020b). Changes in the composition of the gut microbiome in patients with LGI1 or CASPR2 antibody-positive encephalitis were found to correlate with neuroinflammatory states using metagenomic birdshot sequencing and may be mediated by reduced short-chain fatty acid-producing (SCFA) production (Ma et al., 2020). In addition, in another 16sRNA sequencing study against LGI1 encephalitis, ecological disturbances in the gut microbiome of patients with AE were found, along with a significant reduction in some genera of gut microbes with SCFA (Ma et al., 2020). Notably, Hao Chen et al. performed fecal microbiota transplantation (FMT) in mice with anti-NMDAR encephalitis patients and found that the mice developed microbiota deficiencies, were hypersensitive, and cognitively impaired, demonstrating a strong association between gut microbes and AE (Chen et al., 2020).

The results of these studies all indicate that there is a relationship between gut microorganisms and the pathogenesis of AE, in which SCFA plays a non-negligible role, and further in-depth studies are needed to apply it in the diagnosis of AE as well as in clinical treatment.

3.5. Metagenomics

Metagenomics can comprehensively characterize all DNA or RNA present in a sample and thus analyze the microbiome in patient samples as well as the human host genome or transcriptome (Chiu and Miller, 2019). In AE studies, metagenomics has been applied to analyze CFS specimens, serum, and stool samples. This analysis can help distinguish AE from infectious encephalitis by revealing the unique expression of the genome of AE patients at the macro level. In addition, metagenomics analysis may provide new insights into the diagnosis and treatment of AE, especially in the identification of differentially expressed biomarkers. These biomarkers provide potential clinical applications for early diagnosis and targeted therapy of AE.

Infectious encephalitis (IE) and AE have overlapping clinical manifestations, including fever and seizures. However, the treatments for IE and AE are not the same. At the same time, the early diagnosis of AE is complicated, and it is easy to misdiagnose it as IE or to miss the diagnosis because of the negative antibody test and the lack of specificity of other tests. Therefore, it is very important to distinguish between IE and AE in early diagnosis, and more sensitive and effective methods are needed to distinguish between IE and AE.

In recent years, Macro genomes Next Generation Sequencing (mNGS) methods have been widely used for the detection of pathogens in clinical samples (Gu et al., 2019b; Gu et al., 2021) due to their unbiased, comprehensive, and highly sensitive detection of pathogens, including applications for the diagnosis of suspected infectious encephalitis (Wang et al., 2020; Xing et al., 2020; Wilson et al., 2019). Several clinical studies have shown that mNGS is effective in improving the detection of novel, rare, or unexpected pathogens, thereby enhancing the ability to diagnose infectious encephalitis and thus differentiate between AE and IE (Brown et al., 2018).

Macro transcriptome Next Generation Sequencing (mtNGS) is a type of RNA-seq that not only obtains information about the pathogen and microbiome in a sample but also presents host gene expression profiles that reveal host responses (Ramachandran et al., 2022). Several studies have attempted to find markers to differentiate AE from IE in terms of clinical features, but the specificity and sensitivity of these markers need to be further investigated (Michael et al., 2016). It has been reported that mtNGS can screen for specific biomarkers from host and/or pathogen transcriptional profiles to identify specific pathogen infections (de Araujo et al., 2016) to distinguish between different types of pathogen infections or for early diagnostics (Bhattacharya et al., 2017). Siyuan Fan et al. utilized the transcriptomic profile of mtNGS to find biomarkers to distinguish AE from IE. They constructed an efficient classifier consisting of significant DEGs between IEs and AEs by meta-transcriptomic analysis of CSFs, with the potential to differentiate between diseases within 2 days, or even within the next few hours, much faster compared to traditional methods that can take weeks (Venkatesan and Geocadin, 2014; Fan et al., 2023).

In addition to differentiating AE from IE in early diagnosis, metagenomics can be applied to study the presence of viruses in the cerebrospinal fluid of patients with AE, providing insight into the mechanisms by which viruses may induce AE. The etiology of AE is usually unclear, and most findings suggest that viral infections, especially HSV, are associated with its pathogenesis (Morris et al., 2016; Bradshaw et al., 2015). Virus-mediated brain tissue damage may lead to exposure to normally isolated neuronal cell antigens, and “molecular mimicry” of viral proteins is also a possible cause of autoantibody production (Mueller et al., 2018). The most studied virus is herpes simplex virus, which triggers autoimmunity to NMDAR, dopamine D2 receptor (D2R), and gamma-aminobutyric acid type A receptor (GABA-AR) (Dutra et al., 2018; Armangue et al., 2015). In addition, a variety of other viral pathogens, including West Nile virus (WNV), varicella zoster virus (VZV), and even SARS-CoV-2, have been implicated in AE development (Prakash et al., 2019; Karagianni et al., 2019; Panariello et al., 2020; Monti et al., 2020).

Perlejewski K et al. identified the Torque teno virus (TTV) in a patient by metagenetic study and confirmed its presence by specific PCR (Perlejewski et al., 2020). TTV is very common in the general population and is considered an orphan virus. In addition, another NGS-based study reported the presence of TTV in the cerebrospinal fluid of patients with encephalitis/meningitis (Kang et al., 2017; Eibach et al., 2019; Ikuta et al., 2019). Interestingly, TTV was previously associated with human autoimmune diseases such as herpetic pemphigoid and lupus erythematosus (Gergely Jr et al., 2005; Blazsek et al., 2008).

3.6. Radiographic

Most patients with encephalitis receive an MRI of the brain early in the disease. It usually appears normal or nonspecific and sometimes abnormal, which may suggest an autoimmune cause. In contrast, EEG changes are rarely specific. MRI findings in combination with clinical findings can be used to differentiate AE from tumors, ischemic or hemorrhagic infarcts, and infections, and to reduce the rate of misdiagnosis of AE. For different subtypes of AE encephalitis, MRI examination usually shows different abnormal findings with certain specificity, which is useful for clinical diagnosis and early differentiation.

MRI imaging features of limbic system encephalitis include high signal within the proximal mesial temporal lobes (amygdala and hippocampus) on T2-weighted and fluid-attenuated inversion recovery (FLAIR) sequences, although any structure within the limbic system may be involved. Findings may be unilateral at first but can progress to involve both temporal lobes. Associated parenchymal swelling can be seen but eventually subsides and leads to atrophy (mesial temporal sclerosis) over months to years (Urbach et al., 2006). FDG-PET shows increased FDG uptake initially, followed by hypometabolism (Guerin et al., 2019). It is important to note that herpes encephalitis caused by herpes simplex virus is similar to limbic system encephalitis in terms of imaging presentation, showing abnormal distribution of signal within the temporal lobe, insula, and cingulate gyrus. However, HSV encephalitis more often shows diffusion restriction and hemorrhage than limbic system encephalitis (Madhavan et al., 2020). In addition, AE encephalitis is more likely to show basal ganglia involvement on imaging.

A variety of encephalitis, including NMDAR encephalitis, anti-VGCC, anti-GABA-A, and anti-GluR3 encephalitis show involvement of cortical areas outside the limbic system on imaging. Imaging features of this type of encephalitis on MRI include multifocal areas of cortically-based T2/FLAIR hyperintensity with or without pyriform enhancement or subtle enhancement of the overlying meninges (Dalmau and Rosenfeld, 2014; Kelley et al., 2017). As the disease progresses, cortical laminar necrosis may develop. Cortical-based changes in T2/FLAIR signaling are also seen in low-grade gliomas or some metabolic disorders such as hypoglycemic encephalopathy and hyperammonemia encephalopathy (Ball et al., 2022).

Imaging features of anti-CV2, anti-D2 encephalitis, as well as anti-NMDAR encephalitis on MRI, include hyperintense T2/FLAIR abnormalities in the basal ganglia, often bilateral, with sparing of the mesial temporal lobes (Kelley et al., 2017). There is usually no diffusion restriction (Kelley et al., 2017), which helps to distinguish this subgroup of AE from prion diseases (e.g. Creutzfeldt-Jakob disease) as well as ischemic infarction (da Rocha et al., 2015).

There are also rare types of AE encephalitis that involve the cerebellum or brainstem. Follow-up imaging of AE encephalitis involving the cerebellum months to years later may reveal cerebellar atrophy, which typically manifests as ataxia, nystagmus, and speech and swallowing difficulties (Madhavan et al., 2020). Brainstem encephalitis or rhombencephalitis usually involves the brainstem, which refers to inflammation of the brainstem with variable involvement of the cerebellum and cerebellar peduncles, can present with ataxia, speech difficulties, sensorineural hearing loss, ophthalmoplegia, and even central respiratory depression (Jubelt et al., 2011). The acute phase may be normal or show T2/FLAIR hyperintensity within the brainstem, with or without cerebellar involvement. Often, these disorders progress to show atrophy of the brainstem and cerebellum on follow-up imaging (Ball et al., 2022).

4. Multi-omics co-analysis

Currently, multiple genetic loci, differentially expressed non-coding RNAs, protein molecules, and metabolic molecules associated with the pathogenesis of AE have been identified by different histologic analyses. However, no single potential biomarker has been supported by more than two histologic techniques. Although all histological approaches have contributed greatly to the detection of possible AE biomarkers, single-omics cannot provide a comprehensive and systematic analysis of the complex biological mechanisms underlying the pathogenesis and progression of AE. Therefore, the emergence of methods for integrating multi-omics data has provided an opportunity for a holistic picture of AE. In this section, we describe the use of integrated analysis of multi-omics data to facilitate the study of potential biomarkers for AE.

A common multi-omics integration strategy is to obtain tissue samples including serum as well as cerebrospinal fluid after which multiple histological analyses are performed separately and the differentially expressed molecules obtained are integrated as potential biomarkers to enhance the understanding of disease pathogenesis and the specific role of that potential biomarker in different pathways (Fig. 1).

Here, we specifically give two methods of combining analyses. Combining genomic and transcriptomic data can help us understand how genetic variants such as mutations, copy number variations, or single nucleotide polymorphisms (SNPs) affect the development of AE by regulating gene expression. The combination of GWAS and RNA sequencing (RNA-seq) technologies can reveal mutation sites or gene expression differences associated with AE, and further screen for genes that are significant in clinical patient samples changes in clinical patient samples.

For example, Anja K et al. (Tietz et al., 2021) combined GWAS and RNA-seq analyses to screen two immune and neurological function genes associated with AE: one located on chromosome 15 containing only the LRRK1 gene, and the other located on chromosome 11 centered on the ACP2 and NR1H3 genes in a region of higher linkage disequilibrium. These findings are helpful for the early diagnosis, monitoring, and treatment strategy of AE. In addition, Guiyou Liu et al. (Liu et al., 2017) identified several immune-related genes and signaling pathways associated with multiple sclerosis, including antigen presentation and T cell activation, natural killer cell signaling pathway, T cell receptor

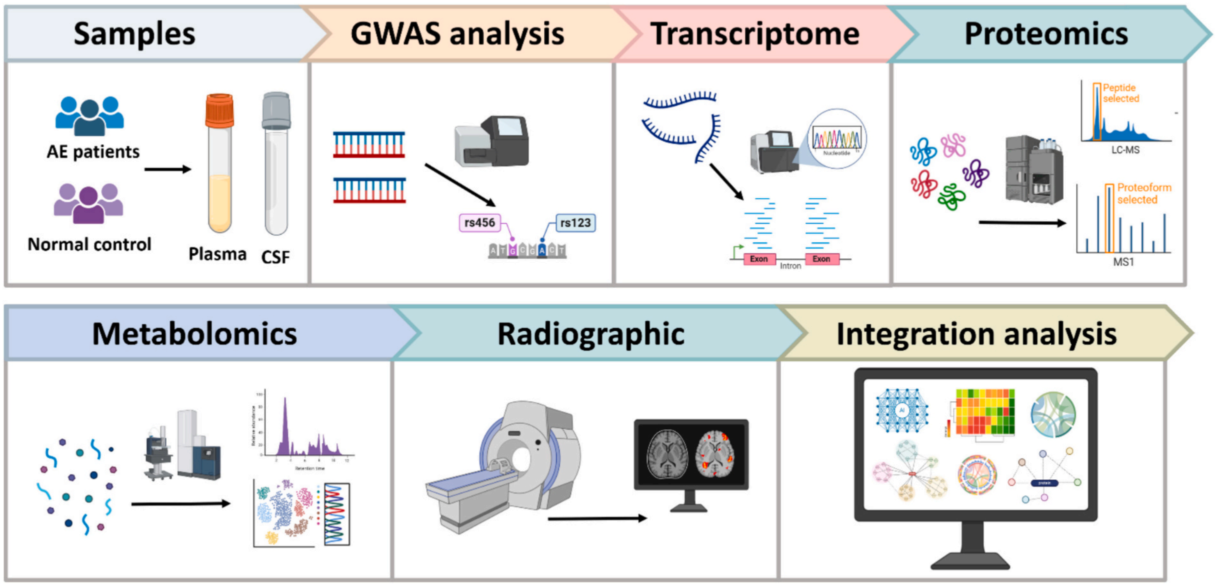

Fig. 1. Schematic diagram of the process of joint multi-omics analysis of potential biomarkers for AE.

signaling pathway, and immunoglobulin production and regulation, by integrating GWAS and RNA-seq data. These studies provide new directions for further pathological studies and targeted therapies.

In addition, combine data from gene expression, protein levels, and metabolomics to further reveal downstream metabolic changes by verifying that up-or-down-regulation of gene expression can be directly reflected in changes in their corresponding proteins. Determine the interaction network of certain protein molecules as well as the upstream and downstream molecular mechanisms, and study their specific mechanisms as potential biomarkers for AE.

Studies combining multi-omics approaches can provide a more comprehensive and systematic biological view of AE. This multilevel and multidimensional data integration provides a solid foundation for the screening and clinical application of potential biomarkers for AE. In the future, with the advancement of technology and the continuous improvement of data integration methods, the comprehensive analysis based on multi-omics data will likely promote the early diagnosis, precision treatment, and individualized medicine of AE.

In addition, multi-omics analysis can be combined with machine learning methods for pattern recognition analysis. By processing and analyzing a large amount of medical data, including images, gene expression, clinical trial data, and clinical features, automated feature extraction, and pattern recognition can be applied to screening and early diagnosis of AE in the future. In recent years, several studies have begun to explore the application of machine learning in imaging to identify potential biomarkers of AE. Manfred et al. (Musigmann et al., 2025) et al. used machine learning based on pre-contrast T2-weighted MR images acquired during symptom onset to predict antibody serostatus in patients with suspected autoimmune encephalitis (AE), which may provide an effective tool for early diagnosis of AE. In immunohistochemical testing, machine learning may identify additional potential diagnostic markers and further improve the diagnostic accuracy and efficiency of AE by analyzing the characteristic expression patterns of immune cells, cytokine levels, and pathology image data in brain tissue sections.

5. Discussion

The diagnosis and treatment of AE have improved dramatically in recent years. However, clinical diagnosis based on antibodies alone has drawbacks and difficulties in clinical application. Different AE subtypes exhibit antibody variability, involving specific antibody markers that are closely associated with patients' clinical symptoms and disease progression, with different pathologic mechanisms and clinical manifestations (Graus et al., 2016). Existing commercially available antibody tests are unable to screen for multiple possible subtypes of AE at the same time, which may result in the underdiagnosis and misdiagnosis of rarer AE subtypes. In addition, there are a large number of seronegative patients who meet the clinical diagnosis of AE but lack detectable antibodies (Gugala et al., 2011). Currently, most antibody-positive tests rely on immunochemical analysis of serum or cerebrospinal fluid samples (including ELISA, WB, etc.), which can easily lead to false-positive or false-negative tests due to differences in sensitivity and specificity of the tests (Dadkhah and Bianciardi, 2016). Therefore, emerging antibody assays or additional multiple markers are needed to guide diagnosis, differentiation of AE subtypes, and prognostic treatment.

Based on the widespread use of multi-omics in AE research, this article summarizes research advances related to the use of different histologies to identify biomarkers, including some specific proteins, miRNAs, or metabolite changes (Fig. 2). However, a single histology does not provide a comprehensive and systematic analysis of the biological complexity of AE. Therefore, a new idea - multi-omics integration analysis for the diagnosis and treatment of AE - has been further proposed. Multi-omics integration can screen for disease-associated genes, proteins, and metabolite changes to provide clues for diagnosis and prognosis determination, opening a new avenue for mining potential biomarkers for autoimmune encephalitis. Similar multi-omics integration strategies have progressed in the study of other immune diseases (Liu et al., 2022), identifying potential markers that support studies related to AE. In MS, multi-omics has identified several potential biomarkers that may play a potential role in the diagnosis and prognosis of MS. These markers include IgG index, anti-AQP-4 antibody, NFL, and CHI3L (Ziemssen et al., 2019). However, we need to note that biomarker validation for MS is slow, with challenges in reproducibility and standardization (Serrano et al., 2024).

In the classic autoimmune disease, systemic lupus erythematosus (SLE), there have been multi-omics studies identifying genetic susceptibility associated with neurological involvement, such as the association of genes like IRF5 and STAT4 with SLE (Abelson et al., 2009; Liu et al., 2024). In addition, antiphospholipid antibodies, neuromarkers, and other markers that may be associated with neurological damage have also been identified (Zandman-Goddard et al., 2007). Metabolic profiling can reveal new potential biomarkers and provide support and new ideas for the study of autoimmune encephalitis. In the

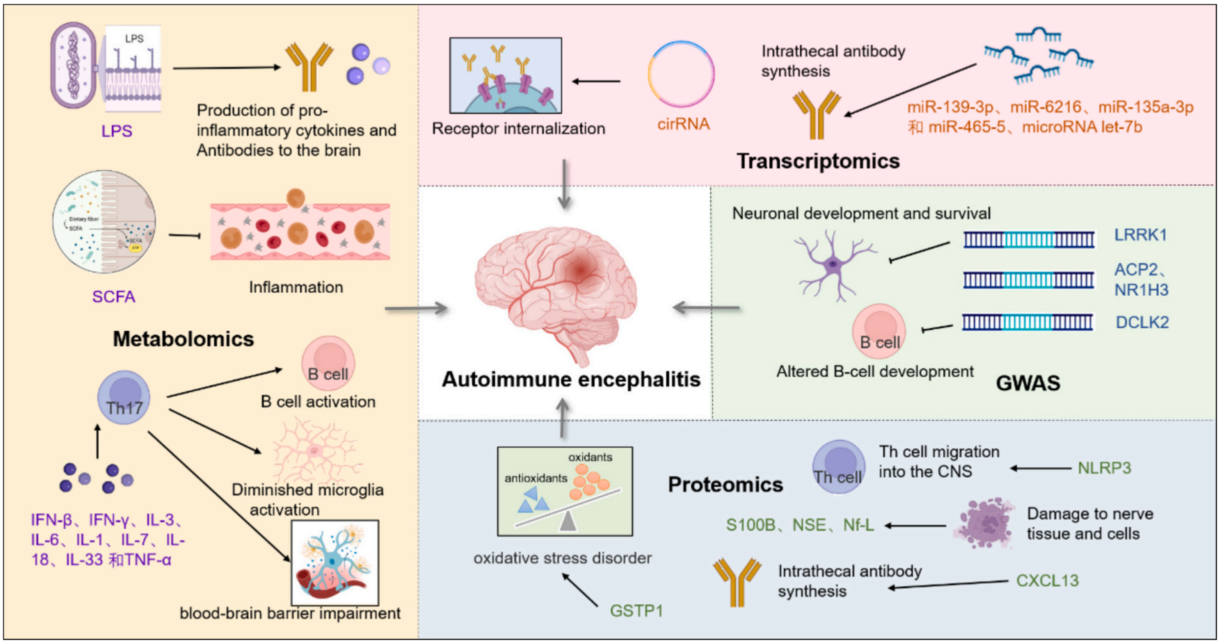

Fig. 2. Potential biomarkers of autoimmune encephalitis and possible mechanisms involved. We list biomarker molecules that are associated with the pathogenesis of AE and potentially of clinical significance by multi-omics studies.

neuroimmune disease, Guillain-Barré syndrome (GBS) (Wang et al., 2022), the more mature application is genomics, and studies have been conducted to identify several genetic variants associated with GBS, especially genes related to immune response (e.g., HLA genes) (Blum et al., 2018). These genetic markers help to reveal susceptibility to GBS.

Previous studies have identified several potential biomarkers for AE, including microRNA let-7b (Weber et al., 2024), CXCL13 (Körtvelyessy et al., 2020), NLRP3 (Knorr et al., 2020), NSE (Loy et al., 2005), S100B (Routsi et al., 2006), Nf-L (Nissen et al., 2021), and others. These emerging biomarkers have a good potential application to represent the presence of actual AE pathological changes in the patients from whom the samples were obtained and are expected to be further studied in depth to determine their potential use as diagnostic markers (Reel et al., 2022). The application of multi-omics in AE is still in its early stages, and its clinical utility remains largely unproven. The translation of potential biomarkers into clinically diagnostic usable biomolecules remains limited. Promoting the development of multi-omics for AE has good prospects and requires overcoming difficulties such as high cost, large-scale multicenter validation, and complexity of data integration and analysis. With technological advances and in-depth research, multi-omics has the great advantage of being able to identify new biomarkers by integrating data from different groups, especially in the early or latent stages of the disease, to help diagnose AE early and predict its clinical progression.

Interestingly, we offer the possibility that gut microbes influence AE through the brain-gut axis. Numerous studies have demonstrated significant gut microbiota disruption and abnormalities in downstream metabolites in patients with AE. Characterization of compositional and functional differences in the gut microbiome using macrogenomics sequencing suggests that changes in the composition of the gut microbiome in patients with AE correlate with neuroinflammatory states and may be mediated by reduced short-chain fatty acid production (SCFA). SCFAs are products of gut microbial metabolism and are produced primarily from dietary fiber fermented by gut flora in the colon. SCFAs are produced in the gut by activating the GPR41 and GPR43 receptors, which are involved in regulating the immune response. They can promote the production of regulatory T cells (Treg) and reduce the activity of Th17 cells, thereby suppressing the autoimmune response and inflammatory process (Furusawa et al., 2013). However, the causal relationship between intestinal flora dysbiosis and the reduction of SCFA and AE is not clear. Future studies rely on large sample data to deeply investigate the mechanism of intestinal flora disorders and metabolite abnormalities in AE, to deeply study the intestinal flora through the brain-intestinal axis in AE, and to further propose microbiome-based diagnostic or therapeutic approaches.

In addition, SCFAs can indirectly influence the inflammatory response in the brain by regulating the composition of the gut microbiota (Baxter and Foss, 2014). Studies have shown that SCFAs can regulate the brain microenvironment through the gut-brain axis and inhibit the over-activation of immune cells (e.g., microglia) in the brain (McCarthy et al., 2020), thereby reducing neuroinflammation and potentially exerting a mitigating effect on diseases such as autoimmune encephalitis. In MS and Parkinson's disease, it has been suggested that gut flora dysbiosis may affect CNS autoimmunity (Doifode et al., 2021). Specifically, microglia collaborate with other cell types (especially astrocytes and oligodendrocytes) to maintain the balance between the functions and responses involved in maintaining brain homeostasis (Malvaso et al., 2023). Together with astrocytes and the blood-brain barrier, they form what is known as the "neurovascular unit," participating in responses to brain injury and thereby coordinating neuroinflammatory processes (Liu et al., 2020). Additionally, they collaborate with oligodendrocytes to regulate the complex process of myelin formation under both physiological and pathological conditions (Kalafatakis and Karagogeos, 2021). Microglia can exert protective effects, but can also be harmful. The classic "M1" pro-inflammatory phenotype is considered neurotoxic, while the "M2" anti-inflammatory phenotype is considered neuroprotective. The M1-M2 polarization of microglia corresponds to distinct active phenotypes associated with harmful inflammatory effects or reparative tasks (Masuda et al., 2019; Molina-Martínez et al., 2020). Previous studies have primarily used immunofluorescence staining to investigate microglial cell types and quantify changes in the expression of related inflammatory factors and chemokines. Currently, multi-omics integration technologies can be utilized to further explore the role of microglia in neuroinflammation. By combining spatial transcriptomics with immunofluorescence staining, it is possible to localize diseased glial cells in specific brain regions while distinguishing between pro-inflammatory (M1) and anti-inflammatory (M2) phenotypes. Single-cell epigenomics can be used to identify chromatin regions that are specifically opened during inflammation. Subsequently, transcriptomics and proteomics technologies can be combined to investigate whether the opened regions are associated with pro-inflammatory genes and whether epigenetic regulation affects protein secretion. Additionally, by integrating multi-omics technologies with machine learning methods for pattern recognition analysis, it is possible to comprehensively study the role of microglia in neuroinflammation and their impact on the pathogenic process of AE.

In recent years, as understanding of AE has deepened, researchers have discovered that its clinical manifestations far exceed the traditional understanding of psychiatric behavioral abnormalities and seizures, with the emergence of more complex and variable new symptoms, including neuropsychological disorders and neuro-respiratory spectrum disorders, encompassing memory disorders, cognitive disorders, executive function disorders, psychomotor disorders, sleep disorders, and abnormal respiratory rhythms, among other multi-system manifestations. For example, LGI/CASPR2-positive patients have exhibited retrograde amnesia (Antonio et al., 2025), IgLON5-positive patients have demonstrated abnormal inspiratory phase respiratory sounds during sleep, as well as dysarthria and dysphagia (Hasan et al., 2025; Yapp et al., 2025), and GABAAR-positive patients have presented with psychomotor retardation, executive dysfunction, emotional regulation abnormalities, and decision-making impairments. Children with anti-NMDAR encephalitis may experience significant regression in language skills, social skills, and academic abilities, often manifesting as acquired neurodevelopmental regression. Recent studies indicate that these children may still exhibit subtle social information processing impairments during the recovery phase, such as delayed facial emotion recognition and reduced eye contact (Wang et al., 2019). Additionally, a recent cohort study of AE patients revealed reduced sleep efficiency, decreased high-frequency coupling (HFC) indicative of deep sleep, elevated proportions of rapid eye movement (REM) sleep and wakefulness, elevated respiratory disturbance indices, and significant autonomic dysfunction (Liao et al., 2024).

The delayed identification and misdiagnosis of new clinical manifestations pose additional challenges for diagnosing AE. Therefore, it is essential to conduct a stratified analysis of patients based on new symptoms, including neuropsychological status and neuro-respiratory spectrum. Additionally, utilizing multi-omics technologies to explore the characteristic biomarkers of new symptoms and the underlying complex pathophysiological mechanisms can provide new insights into AE diagnosis. Previous studies have shown that the CSF biomarker neurofilament light chain protein (NfL) reflects neuronal axonal damage and is associated with long-term cognitive outcomes (Taraschenko et al., 2024); fMRI studies have demonstrated that weakened functional connectivity between the hippocampus and the default mode network (DMN) is significantly associated with memory impairments (Peer et al., 2017). The application of multi-omics has identified additional characteristic biomarkers, opening up new possibilities for research into the pathophysiological mechanisms of AE (Liao et al., 2024). Additionally, machine learning can be used for intelligent analysis of neuropsychological data to classify AE patients, distinguishing between anti-NMDAR, anti-LGI1, and anti-GABAA receptor encephalitis. Developing various machine learning models that integrate clinical data, clinical scores, and laboratory indicators can be used to predict recovery outcomes.

Author contribution

Mingjing Zhao: Literature review, Critical discussion, Manuscript drafting; LiWen Jin: Manuscript drafting. Jing Li modify the article. Zhaohui Luo designed the outline and revised the manuscript; All the authors have approved of the publication of the manuscript. All authors reviewed and approved the final manuscript.

CRediT authorship contribution statement

Mingjing Zhao: Writing – original draft, Visualization, Data curation, Conceptualization. **Jing Li:** Supervision. **LiWen Jin:** Writing – review & editing. **Zhaohui Luo:** Supervision, Conceptualization.

Declaration of competing interest

None declared.

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Data availability

No data was used for the research described in the article.

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